

General Troubleshooting Guide — Process Line 2

1. Purpose

This guide provides a quick-reference troubleshooting matrix for the most common abnormal conditions on Process Line 2 rotating equipment, cross-referencing symptoms to likely causes and the manual section with more detail.

2. Symptom: Pump P-102 Bearing Temperature High

Likely causes, in order of frequency observed in the last 12 months of maintenance records:

- Low cooling water pressure in the shared cooling loop (see compressor_manual.pdf, section 3, for the interaction with compressor C-201) — most common cause.
- Low or degraded lubrication oil — see pump_manual.pdf, section 4.
- Pump operating far from Best Efficiency Point — see pump_manual.pdf, section 4.
- Bearing wear (typically only after 18+ months since last replacement) — see pump_manual.pdf, section 5, for inspection sequence.

3. Symptom: Compressor C-201 Discharge Temperature High

- Cooling water flow restriction — check shared loop, same as pump P-102 cooling issues.
- Worn or leaking suction/discharge valves — see compressor_manual.pdf, section 4.
- Cylinder fouling on the cooling water side.

4. Symptom: Simultaneous High Temperature on Pump and Compressor

When both P-102 and C-201 show high temperature readings at the same time, always check the shared cooling water loop FIRST (strainer, cooling water circulation pump, isolation valves) before investigating either machine individually. This is the most time-efficient diagnostic path and has resolved the majority of dual-alarm events historically.

5. Escalation

If the cooling water loop is confirmed normal and temperatures remain high, follow the shutdown criteria in safety_manual.pdf, section 4, and raise a maintenance work order per maintenance_procedures.pdf, section 4.